

IMPLEMENTATION OF BIOPORE AND SOIL FECAL BACTERIA FATE AND TRANSPORT ROUTINES IN THE ROOT ZONE WATER QUALITY MODEL (RZWQM)

J. A. Guzman, G. A. Fox

ABSTRACT. Surface runoff and infiltration are primary fecal bacteria transport mechanisms occurring in agricultural fields following manure application. The presence of macropores and biopores notably increases water infiltration, decreases surface runoff, and allows fecal bacteria and water to rapidly bypass the soil matrix during rainfall or irrigation events. This research incorporated fecal bacteria transport and biopore routines into the Root Zone Water Quality Model (RZWQM) to simulate flow and fecal bacteria transport through the soil, to subsurface drainage, and in runoff. Flow, soil fecal bacteria transport routines, and the influence of macropores and biopores on *E. coli* transport were evaluated based on a series of soil column experiments (28 × 50 × 85 cm) with two contrasting soil types (loamy sand and sandy loam) in which artificial, open surface connected biopores were created at two different lengths (55 cm and 65 cm). In addition, simulation of fecal bacteria transport in runoff was evaluated from a series of 2 × 2 m plot experiments treated with poultry litter and subjected to two rainfall storm event intensities. The new routines improved RZWQM's capability to predict rapid flow (e.g., shape of the hydrograph, time to peak, and flow breakthrough; Nash-Sutcliffe efficiency index ranged between 0.65 and 0.75 for model validation) and soil fecal bacteria transport, e.g., absolute error for the simulated event mean concentrations (EMCs) ranged between 4% and 109%. Fecal bacteria transport concentrations were underestimated in some cases, but the modified model captured the trend in concentrations observed during the soil column and plot experiments. The updated model is a simple prediction tool capable of simulating fecal bacteria transport in runoff and subsurface drainage with and without the presence of biopores.

Keywords. Bacteria transport, Biopores, *Escherichia coli*, Fecal bacteria, RZWQM, Soil fecal bacteria modeling.

Fecal bacteria concentrations in surface water are commonly reported as a significant cause of water quality degradation throughout the world (Gale, 1996; Haack et al., 2003; Dorner et al., 2006). Animal feces contain various coliforms such as *Escherichia coli* (*E. coli*), including the harmful pathogen *E. coli* O157, that can contaminate rivers, lakes, and groundwater when animal waste is land-applied (McGeachan and Vinten, 2003). Surface runoff and water infiltration are primary fecal bacteria transport mechanisms following manure application in agricultural fields (Jamieson et al., 2002; Tyrrel and Quinton, 2003). During infiltration, the physical (e.g., soil texture, macroporosity, and organic matter), chemical (e.g., pH, temperature, soil minerals and amorphous metals, soil solution ionic strength, and exchangeable ion concentration), and biological (e.g., microbial population dynamics) interactions occurring in the soil matrix may result in fecal bacteria

inactivation, transport retardation, or immobilization (Fontes et al., 1991; Unc and Goss, 2003; Darnault et al., 2004; Pachepsky et al., 2006). In most cases, these variables and processes are simplified when attempting to analyze soil fecal bacteria fate and transport due to complexity, specificity, lack of knowledge, and uncertainty.

The presence of macropores (i.e., soil fractures, mesopores, and biopores) in soils may increase infiltration and interflow, allowing rapid transport of solutes and microorganisms to deeper soils and/or subsurface drainage systems (Kemper et al., 1988; Shipitalo and Gibbs, 2000; Akay and Fox, 2007; Guzman et al., 2009). Biopores are macropores formed by biological activity that may be different from macropores created by soil structure and fractures due to their unique geometry and origin. Biopores act like pipes, rapidly propagating wetting fronts and in some cases interconnecting the surface to drainage systems.

A few physically based or conceptual hydrological models at the column, plot, and catchment scale have been used to simulate soil bacteria transport and macropore-facilitated flow and transport. Examples of these models are MACRO (Jarvis, 1994) and HYDRUS (Šimunek et al., 2005, 2006). Recently, several articles have been published addressing the importance of transport driven by preferential flow in the vadose zone; for instance, Köhne et al. (2009a, 2009b) published a detailed review of model applications for structured soils and transport of tracers and pesticides, and Allaire et al. (2009) reviewed laboratory and field techniques for evaluating preferential flow in soils at different scales. In most cases,

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bacteria transport has been simulated in a continuum porous media representation by solving the advection-diffusion equation in one, two, or three dimensions based on flow computations using Richard's equation and/or the Green-Ampt method. Soil structure has been incorporated into models by assuming a uniform soil macroporosity (i.e., fractures, macropores, mesopores, and micropores) in dual porosity or dual permeability models. However, assumptions made to simulate water flow at the pore scale in physically based models are too restrictive in the presence of biopores that create lateral differences in water pressure and solute/bacteria concentrations in the unsaturated zone (i.e., physical non-equilibrium) (Jarvis, 2007).

The Root Zone Water Quality Model (RZWQM; <http://ars.usda.gov/Main/docs.htm?docid=17740>; Ahuja et al., 2000) is a one-dimensional, physically based, lumped model with the capability to simulate the physical, biological, and chemical processes occurring in the vadose zone under different scenarios and management practices. RZWQM simulates a one-dimensional vertical soil profile with an integrated subsurface drainage component. Therefore, the model assumes spatial homogeneity and has been typically applied for field-scale applications (Fox et al., 2004; Fox et al., 2006). The main hydrological processes in the vadose zone are modeled using the one-dimensional Richard's equation for soil water redistribution, the Green-Ampt method for infiltration, and the integrated Hooghoudt's equation (Bouwer and van Schilfgaarde, 1963) for subsurface drain flow. RZWQM also has the capability to incorporate soil structure by using uniform macroporosity per soil horizon (Ahuja et al., 1995; Ahuja et al., 2000). Moreover, flow and solute transport from directly connected macropores to subsurface drainage systems can be simulated using the express fraction concept (Fox et al., 2004; Fox et al., 2007). This model has been considered for use by governmental agencies (e.g., the U.S. Environmental Protection Agency) for simulating contaminant transport in soils; however, the model currently lacks routines for soil bacteria transport and facilitated flow and transport in the presence of biopores. During the infiltration process, biopores allow water and contaminants to rapidly bypass specific soil depths (Akay et al., 2008). This condition allows the propagation of a secondary wetting front to the end of the biopore length and along the biopore. A rapid rise in the water table may result from biopore flow bypassing a portion of the soil matrix, increasing flow and transport at the drain as a function of the soil properties, macropores, and layer thickness between the end of the biopore and the subsurface drain (Akay and Fox, 2007; Akay et al., 2008; Guzman et al., 2009).

Biopores such as those created by anecic earthworms are unique macropores that are critically important for understanding rapid increases in drainage flow and transport during irrigation and rainfall events following manure applications. In many cases, a biopore concept may be required to expand RZWQM's capabilities to better predict fecal bacteria fate and transport at the field scale, where fecal bacteria fate and transport in the soil is largely controlled by macropore processes (Garbrecht et al., 2009; Guzman et al., 2009). Therefore, the objectives of this research were to (1) incorporate routines in RZWQM to simulate fecal bacteria transport through the soil matrix, macropores, and in surface runoff; and (2) to incorporate an open surface connected (OSC) biopore routine to improve flow and fecal

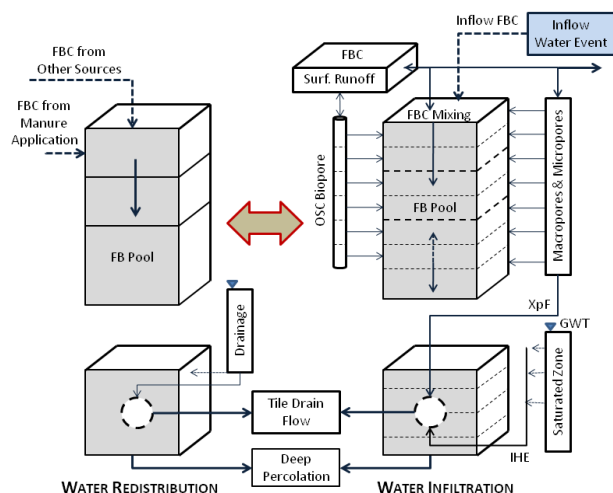


Figure 1. Conceptual diagram of the main RZWQM hydrological processes and the new biopore concept: FBC is fecal bacteria concentration, FB is fecal bacteria mass, XpF is the express fraction, OSC is open surface connected macropore, IHE is the integrated Hooghoudt's equation, and GWT is the groundwater table.

bacteria transport simulations to subsurface drains. Few, if any, models currently exist for simulating fecal bacteria fate and transport through subsurface drained soils and incorporate soil macropore, mesopore, and biopore flow.

MATERIAL AND METHODOLOGY

Routines to simulate fecal bacteria fate (i.e., die-off) and transport (i.e., convection, straining, attachment, and detachment) were incorporated into RZWQM's water redistribution and infiltration subroutines. Additionally, a routine simulating OSC biopores from anecic earthworms was added to the infiltration module for rapid flow and transport to a specific soil depth (fig. 1). No additional routines were included to account for surface runoff or subsurface drain flow routing.

FECAL BACTERIA FATE AND TRANSPORT MODELING

Soil fecal bacteria fate and transport were modeled based on the conservation of mass at the layer discretization during infiltration or at the horizon discretization during redistribution in a unitary volume (e.g., 1 cm³ fecal bacteria pools). Fecal bacteria transport mechanisms such as advection, attachment, detachment, straining, and die-off were considered in the liquid and/or solid phase. Attachment was represented by a non-linear function (Guzman et al., 2012), while detachment was assumed to occur linearly. Straining was assumed linearly proportional to the liquid phase concentration (Foppen et al., 2005). The fecal bacteria mass transported in the liquid and solid phases from an adjacent layer were given by the following equations:

$$\left(\frac{\Delta N}{\Delta t} \right)_{LP} = \left(\frac{u}{\Delta z} C \right)_{adv} - \left(\frac{u}{\Delta z} K_{str} C \right)_{str} - \left(\frac{\rho_{bulk}}{\Delta t} q \right)_{att} - \left(\frac{N}{\Delta t} e^{-kt} \right)_{dieoff} \quad (1)$$

$$\left(\frac{\Delta N}{\Delta t}\right)_{SP} = \left(\frac{u}{\Delta z} K_{str} C\right)_{str} + \left(\frac{\rho_{bulk}}{\Delta t} q\right)_{att} - \left(\frac{D_{det}}{\Delta t} N\right)_{det} - \left(\frac{N}{\Delta t} e^{-kt}\right)_{dieoff} \quad (2)$$

$$q = AC^b \quad (3)$$

where C is the fecal bacteria concentration in the liquid phase (MPN mL⁻¹ or CFU mL⁻¹); u is the flow velocity at the receiving layer (cm h⁻¹); ρ_{bulk} is the soil bulk density (g cm⁻³); q is the attached fecal bacteria concentration (MPN g⁻¹ or CFU g⁻¹); A and b are constants for the nonlinear attachment function; K_{str} is the straining coefficient; D_{det} is the detachment coefficient; Δt is the time step (h); Δz represents the 1 cm or horizon soil layer thickness during infiltration or water redistribution, respectively; k is the die-off rate (h⁻¹); N is the fecal bacteria mass in the liquid or solid phase pools per unitary volume (MPN cm⁻³ or CFU cm⁻³); and *adv*, *str*, *att*, *LP*, and *SP* are abbreviations for advection, straining, attachment, liquid phase, and solid phase, respectively. Note that either CFU or MPN units can be used consistently; enumeration techniques have intrinsic variability and uncertainty along with practical limitations (Gronewold and Wolpert, 2008).

Fecal bacteria die-off was simulated by an exponential inactivation model (Chick, 1908). During the simulation, the first-order die-off rate was adjusted at each time step as a function of the water temperature in the soil profile using the Arrhenius equation (Hrshikesh et al., 2007; Garzio-Hadzick et al., 2010):

$$k = A_r e^{-\frac{E}{RT}} \quad (4)$$

where A_r is a constant in the Arrhenius equation, E is the activation energy, R is the gas constant, and T is the layer's water temperature. Other variables affecting bacteria fate (e.g., moisture content, microbial competition, UV radiation, pH, and soil organic matter content) were not considered in this version of the model.

In cases in which the integrated Hooghoudt's equation contributed flow to the drain, transport of fecal bacteria was estimated as a function of the advection, detachment, and straining processes (eqs. 1 and 2) occurring in the soil volume fraction of the layer separating two drain pipes:

$$N_{DP} = K_{HVF} DP_{SEP} (1 - K_{str}) u C \Delta t \Delta z \Delta y \quad (5)$$

where N_{DP} is the bacteria mass transported to the drain pipe (MPN or CFU), K_{HVF} is the effective horizontal layer volume contributing fecal bacteria, DP_{SEP} is the drain pipe separation distance (cm), K_{str} is the straining coefficient, u is the horizontal flow velocity (cm h⁻¹), C is the fecal bacteria concentration in the liquid phase, Δt is the time step (h), and Δz and Δy are the layer thickness and depth (cm), respectively.

BIPORE FLOW AND TRANSPORT

Biopores were described in the model by a vertical, circular opening reaching a specific soil depth in contact with saturated or unsaturated soil layers (fig. 1). Biopores were assumed to occur homogenously in the one-dimensional soil profile volume as a function of the biopore macroporosity (i.e., biopore volume per 1 cm soil layer volume). Seasonal

biopore variation (e.g., change in the biopore surface opening or clogging) as well as tillage biopore disruption (Ramirez et al., 2009) were allowed by using a daily fraction biopore opening parameter, although not changed in the laboratory and field applications described below. The effective biopore length was expressed as a function of the biopore tortuosity. Flow routed through biopores was proportional to the remaining surface runoff following soil matrix and macropore infiltration (including the express fraction). Biopores propagated water into the soil matrix and upward at the end of the biopore. Note that biopore flow (i.e., water propagating from the biopore burrow to the soil profile) only occurs in RZWQM during the infiltration stage. Transmission of flow from the biopore to each 1 cm layer was estimated as a function of the soil layer water content: (1) for saturated conditions, the minimum between the saturated hydraulic conductivity and the biopore infiltration capacity; and (2) for unsaturated conditions, the minimum between the biopore infiltration capacity and the saturation water deficit. Biopore flow under unsaturated soil conditions occurs when the biopore burrow is filled with water and the adjacent layer's soil moisture content is below saturation.

The theoretical biopore infiltration capacity was estimated by the Hagen-Poiseuille law (Sutera and Skalak, 1993) or tank model for unsaturated or saturated layer conditions, respectively. The actual biopore flow resulted from multiplying the former value by an effective lateral infiltration constant (eqs. 6 and 7) and the daily biopore opening parameter:

$$Q = K_{BPF} K_{eff}^u \frac{\pi r^4 \Delta P}{8 \mu \tau L} \quad (6)$$

$$Q = K_{BPF} K_{eff}^s 2 \pi r \sqrt{2 g L} \quad (7)$$

where K_{BPF} is the biopore opening fraction; K_{eff}^u and K_{eff}^s are the effective lateral infiltration coefficient per soil horizon for unsaturated and saturated conditions, respectively; r is the biopore radius; ΔP is the pressure difference at the corresponding layer; μ is the water dynamic viscosity; L is the biopore soil profile depth at each layer; and τ is the tortuosity. Note that the biopore tortuosity was included in the biopore concept to improve model calibration based on the fact that field observations have reported biopore sinuosity (Shipitalo and Gibbs, 2000). It should be noted that field measurement of this tortuosity is difficult without destructive investigations.

The volume of water propagating downward was limited by the infiltration capacity of the layer containing the end of the biopore. In cases in which the layer at the end of the biopore become saturated, a downward piston flow was allowed until the water front reached the drain pipe depth or the propagating water become exhausted in the adjacent layers. Additional water reaching the drain pipe was added to the subsurface drain flow. The volume of the upward propagating water was the result of the difference between the biopore capacity (i.e., eqs. 6 or 7) and the downward water volume at each time step. Piston flow was also allowed in saturated layers as the water propagated from the end of the biopore to the surface. At the surface, excess water, if any, after the downward and upward propagation processes was returned to surface runoff.

In summary, the biopore concept changed the way in which water and fecal bacteria propagated through the soil profile in terms of (1) allowing flow and concentrations to bypass a group of soil layers as a function of the biopore length; (2) incorporating partial horizontal contributions of fecal bacteria from a specific layer in Hooghoudt's equation, which differed from the RZWQM transport routines that contribute equal concentrations along the layer length; (3) introducing straining and die-off in addition to attachment/detachment and advective transport in that the RZWQM transport components were developed based on liquid phase concentrations while the biopore concept was developed from equilibrated liquid and solid phases.

SOURCES OF FECAL BACTERIA

Three sources of fecal bacteria were considered in the model: (1) fecal bacteria from manure application, (2) fecal bacteria from other sources, and (3) fecal bacteria from inflow water events (fig. 1). Fecal bacteria from other sources represented bacteria mass deposited on the top layer of the soil from animal excretions, such as from rodents and birds. It was included as part of the solid phase bacteria mass based on user-defined monthly mean concentrations and standard deviations. Routines to allow a source of fecal bacteria as part of RZWQM's management practices were added based on the application rate and fecal bacteria concentration of injected liquid manure or broadcasted manure. Following manure application, fecal bacteria was subjected to attachment at the corresponding layer (eq. 2). To allow the model to simulate the laboratory soil column experiments (Guzman et al., 2009), fecal bacteria concentrations in inflow water were allowed.

MODEL CALIBRATION AND VALIDATION

The updated model was evaluated based on its ability to predict the hydrological response (e.g., subsurface drain flow and surface runoff) and fecal bacteria transport observed during two sets of experiments. First, this research used data from a series of five-day experiments carried out in a laboratory soil column (28 × 50 × 85 cm) using an artificial biopore (OSC biopore, 55 cm and 65 cm depths) in two soil types (table 1). In these experiments, the soil column was initially packed and then flushed with distilled water, 48 h later flushed with diluted liquid swine manure (initial *E. coli* concentrations ranging between 8,355 and 15,362 MPN per 100 mL for the cases simulated in this research and reported by Guzman et al., 2009), and then finally flushed with distilled water 48 h after the manure application (Guzman et al., 2009). Flow and *E. coli* concentrations were measured in the subsurface drainage. Note that *E. coli* concentrations were initially undetectable at the subsurface drain following the initial flush for all experiments (Guzman et al., 2009), and *E. coli* concentrations in soil samples were negligible during the column packing. Models were manually calibrated (e.g., soil hydraulic, macropore, and biopore properties) based on the absolute error (i.e., lower than 6%) between the simulated and observed cumulative flow during the manure flush and validated based on the distilled water flush at two experimental times: (1) at the end of the infiltration process, and (2) 9 to 12 h after the manure or distilled water flush initialization. Note that the swine effluent flush and final distilled water flush provided two different flow and transport

Table 1. Soil textural properties for the soil column and plot experiments.

Experiment Type	Soil Type	Bulk Density (g cm ⁻³)	Porosity (%)	Sand (%)	Silt (%)	Clay (%)
Laboratory	Loamy sand	1.6	39.6	85	10	5
Laboratory	Sandy loam	1.4	47.2	65	25	10
Plot	Silt loam	1.4	47.2	20	65	15

conditions to evaluate the modified RZWQM. The swine effluent flush accounted for fecal bacteria transport without background concentrations. On the other hand, the final distilled water flush accounted for the effect of water redistribution and transport with initial background concentrations at different layers within the soil profile (Guzman et al., 2009).

Attachment and detachment parameters affecting transport were obtained from soil sorption experiments after mixing 6 g of soil and diluted swine manure in a series of batch experiments (Guzman et al., 2012). The straining coefficient for the loamy sand was obtained from Garbrecht et al. (2009) but assumed for the sandy loam. The initial values of die-off rate and the effective horizontal soil layer contribution fraction were assumed in all cases. Transport parameters were calibrated (e.g., K_{HVF} and die-off rate in all cases, K_{str} and D_{det} for the silt loam) based on fecal bacteria concentrations observed specifically during the inflow events (i.e., during the infiltration phase only) of the manure flushes, as opposed to both the inflow and redistribution phases. No calibration was performed for the distilled water flush. It was observed that specific, independent calibrations of the model for each case were necessary to obtain optimal model simulations when considering separately the infiltration and water redistribution portions of the experiments. This was hypothesized to be due to the model shifting from infiltration to redistribution computations immediately after the inflow water event, which typically resulted in an immediate underestimation of fecal bacteria concentrations. This phenomenon was also observed by Fox et al. (2007) when simulating pesticide transport to subsurface drains in macroporous soils. For the model simulations presented herein, calibrated fecal bacteria transport parameters from the infiltration phase were used for the redistribution phase simulations.

Second, the model was evaluated using data from a series of experiments using two artificial rainfall events (two-year and five-year return period storm intensities) applied on a 2 × 2 m pastureland plots (silt loam soil, table 1) manually treated with poultry litter at the rate of 4,942 kg ha⁻¹ (Guzman et al., 2010). Average *E. coli* loads applied to the plots were approximately 8.2 × 10¹⁰ MPN ha⁻¹ or 3.3 × 10⁷ MPN. Surface runoff and *E. coli* concentrations were measured at the collection flume located at the drainage end of the plots. Calibration (e.g., soil hydraulic and macropore properties) was manually conducted based on the observed and simulated cumulative surface runoff absolute error (i.e., error lower than 6% was used as a metric for successful calibration) in the two-year (i.e., 58.4 mm h⁻¹) return period event and validated in the five-year (i.e., 76.2 mm h⁻¹) return period event. Note that biopores were not considered for the plot experiments, and the straining and detachment parameters were subjected to calibration based on observed and simulated fecal bacteria concentrations (table 2). The die-off parameters were irrelevant during the simulations due to the short experiment duration. In addition, the calibrated two-year storm event was

Table 2. Soil hydraulic properties for the soil column and plot experiments after calibration.

Soil Type	Experiment	Soil Hydraulic Properties ^[a]						
		h_b	λ	N	K_{sat}	θ_r	θ_s	K_{sat-d}
Loamy sand	OSC-55	7.9	0.48	3.42	11.1	0.035	0.396	0.7
Loamy sand	OSC-65	10.1	0.47	3.42	3.9	0.035	0.396	2.5
Sandy loam	OSC-55	12.7	0.32	2.97	2.6	0.041	0.472	1.0
Silt loam	Plot	7.5	0.36	2.63	0.4	0.015	0.472	15.0

^[a] h_b = bubbling pressure (cm), λ = pore size distribution index, N = exponent for the $k(h)$ curve, K_{sat} = saturated hydraulic conductivity (cm h⁻¹), θ_r = residual water content, θ_s = saturation water content, and K_{sat-d} = lateral saturated hydraulic conductivity to the drain (cm h⁻¹).

used to estimate the equivalent *E. coli* inflow water concentration proportional to the soil *E. coli* background concentration (i.e., fecal bacteria from other sources) observed during the experiments (Guzman et al., 2010). Note that this background concentration was modeled as part of the inflow fecal bacteria concentrations rather than as part of the soil fecal concentrations located in the top two 1 cm layers (i.e., extent of the mixing layer in RZWQM) due to the lack of field measurements. These fecal bacteria background concentrations were hypothesized to be from bird, small mammal, and rodent excretions (Guzman et al., 2010), similar to the poultry litter application, as opposed to soil profile concentrations; further information was not available from the collected data. This proportional fecal background concentration was used as part of the inflow water *E. coli* concentration and resulted from manual adjustments of the inflow water *E. coli* concentration so that the model matched the reported EMC of 6.8×10^3 MPN per 100 mL from control plots (no litter application) in Guzman et al. (2010).

In all cases, the hydrological responses were evaluated by the Nash-Sutcliffe (Nash and Sutcliffe, 1970) efficiency index (NSI), the absolute error between the observed and simulated mass of subsurface drainage or runoff, and the normalized objective function (NOF) (Pennell et al., 1990; Hession et al., 1994; Fox et al., 2006):

$$NSI = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2} \quad (8)$$

$$NOF = \frac{STDD}{\bar{y}} = \frac{\sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}}}{\bar{y}} \quad (9)$$

where STDD is the standard deviation of differences; y_i and $\hat{y}_i$ are the observed and simulated values at time i , respective-

ly; n is the number of observations; and $\bar{y}_i$ is the overall mean of the observed measurements. Fecal bacteria transport was evaluated by the absolute difference between the observed and simulated fecal bacteria event mean concentrations (EMC) (Garbrecht et al., 2009; Guzman et al., 2010):

$$EMC = \frac{\sum \Delta t_i Q_i C_i}{\sum \Delta t_i Q_i} \quad (10)$$

where Δt_i is the time step between two data observations, Q_i is the observed or simulated runoff for the soil column or plot experiments at time i , and C_i is the effluent *E. coli* concentration at time i . The model's ability to predict fecal bacteria transport was also evaluated based on the absolute error in fecal bacteria loads and the NOF computed from observed and simulated fecal bacteria concentrations in subsurface drainage or runoff.

RESULTS AND DISCUSSION

SUBSURFACE HYDROLOGICAL RESPONSE

For modeling drainage flow in the column experiments, the parameters were grouped as a function of the simulation domain and importance during calibration. First, important parameters for the soil matrix domain included the bubbling pressure head, saturated hydraulic conductivity (K_{sat} and K_{sat-d}), and the pore size distribution index (table 2). Second, the fraction of dead-end pores and soil structure macroporosity in the macropore domain (table 3) played a major role in simulating the observed rapid flow in the drain pipe. Macropores served to regulate infiltration fluxes and rapidly increase moisture content in subsequent layers during the primary wetting front (i.e., downward water movement from the surface during infiltration) propagation or as a dual-porosity model during water redistribution. Third, the effective lateral infiltration coefficient (K_{eff}) in the biopore domain controlled water fluxes from the biopore at different soil depths (table 3). Note that all other biopore parameters were measured and unchanged during calibration. Although

Table 3. Soil macropores and biopore properties for the soil column and plot experiments after calibration.

Experiment	Soil Type	Biopore		Macropore Radius (cm)	Macroporosity		FDEP ^[a]	ELWT ^[a]	K_{eff} ^[b]	
		Depth (cm)	Radius (cm)		Biopore (cm ³ cm ⁻³)	Macropore (cm ³ cm ⁻³)			Unsat.	Sat.
Soil column	Loamy sand	55	0.4	0.04	3.59×10^{-4}	6.30×10^{-4}	0.5000	1.0	5.5×10^{-3}	3.5×10^{-2}
	Loamy sand	65	0.4	0.04	3.59×10^{-4}	1.15×10^{-4}	0.4600	1.0	9.5×10^{-5}	5.6×10^{-3}
	Sandy loam	55	0.4	0.04	3.59×10^{-4}	4.30×10^{-5}	0.4647	1.0	1.0×10^{-2}	9.3×10^{-4}
Plot	Silt loam	--	--	0.04	--	3.00×10^{-6}	0.4600	1.0	9.5×10^{-5}	5.6×10^{-3}
	Silt loam	--	--	0.04	--	3.00×10^{-6}	0.4600	1.0	9.5×10^{-5}	5.6×10^{-3}

^[a] RZWQM parameters: FDEP is the fraction of dead-end pores, and ELWT is the effective lateral wetting thickness in RZWQM.

^[b] K_{eff} is the effective biopore infiltration constant.

Table 4. Subsurface drainage soil column calibration and validation results.

Experiment	Flush	Time ^[a]		Inflow (cm ³)	Results ^[b]						
					MC _{obs} (cm ³)	MC _{sim} (cm ³)	NSI	NOF	MC _{err} (%)	f_{avg} (cm ³ s ⁻¹)	PIWEC (MPN mL ⁻¹)
Loamy sand OSC-55	Manure	Start	9:25		0	0			-		
		Inflow	9:45	13,830	1,774	1,779	0.01	0.88	0.3	11.51	153.6
		End	23:07		15,234	14,614			4.1		
	Distilled water	Start	9:45		0	0			-		
		Inflow	9:59	13,920	1,557	2,001	0.65	1.15	28.5	16.05	0
		End	18:24		12,438	11,043			11.2		
Loamy sand OSC-65	Manure	Start	10:40		0	0			-		
		Inflow	12:29	14,190	7,645	7,326	0.76	0.46	4.2	2.15	115.2
		End	22:05		13,203	13,270			0.5		
	Distilled water	Start	9:35		0	0			-		
		Inflow	10:39	14,470	6,349	6,070	0.75	0.74	4.4	3.76	0
		End	23:05		12,440	12,539			0.8		
Sandy loam OSC-55	Manure	Start	11:59		0	0			-		
		Inflow	14:07	14,085	6,666	7,600	0.78	0.40	14.0	1.82	83.6
		End	23:22		10,964	10,366			5.5		

[a] “Inflow” indicates the final time for the inflow event, and “End” indicates the final time for the mass transport assessment.

[b] MC_{obs} and MC_{sim} are the mass curve values for the observed and simulated drainage runoff, NSI is the Nash-Sutcliffe efficiency index, NOF is the normalized objective function, MC_{err} is the mass curve error, f_{avg} is the average infiltration rate, and PIWEC is the proportional inflow water *E. coli* concentration.

biopores are typically grouped in the macropore domain, biopores were classified as a specific domain due to their capacity to rapidly propagate a secondary wetting front laterally along the biopore conduit and vertically below the biopore end, bypassing the primary wetting front propagation. Further global uncertainty and sensitivity analyses are recommended for the updated model but are outside the scope of this article.

During calibration and validation, the biopore routine notably improved drainage flow simulations in the soil column experiments (e.g., NSI in the range of 0.01 and 0.78, NOF in the range of 0.40 to 1.15, and mass curve errors in the range of 0.30% to 29%; table 4). In general, the updated model had the capability to represent the rapid increase in subsurface flow typically observed in the presence of biopores, the hydrograph shape, the breakthrough time, and the time to peak (fig. 2). On the other hand, the simulations without the biopore concept could not model the rapid hydrological response at the drain even though macropores (e.g., from soil structure) were considered during the simulations.

However, note in table 4 that although the validated NSI and absolute errors were within acceptable ranges, the validated NOF indicated the presence of large deviation between the observed and simulated data (e.g., 1.15 and 0.74). These results were explained by improper drainage flow simulation occurring in some cases at the beginning of the inflow event and in all cases right after the inflow event. In figure 2, the inflow event is represented by dashed vertical lines. At the beginning of the inflow event, difference between the observed and simulated flows was explained by the presence of direct macropore connectivity between the end of the biopore and the subsurface drain pipe (fig. 2). On the other hand, following the end of the inflow event, this phenomenon was explained by a rapid decrease in the simulated water table and quick change in moisture conditions at the time the model shifted from the Green-Ampt method simulation in a 1 cm soil layer representation (i.e., infiltration) to Richard's equa-

tion in a soil horizon simulation (i.e., water redistribution) (fig. 1). The effect of this drainage flow underestimation becomes evident in the mass curve of the soil column simulations following the end of the inflow water event (fig. 2). An underestimation of the macropore and biopore depletion was hypothesized to play an important role right after the end of the inflow event. In the observed drainage flow (fig. 2), note that depletion (e.g., subsurface runoff typically characterized by exponential decay with one or more parameters; Chapman, 1999; Gonzales et al., 2009) could be represented by two linear functions on a logarithmic scale following the end of the inflow event. In most cases, with the exception of the SL-OSC-55 experiment, the final portion of the depletion flow (e.g., the second linear function in fig. 2) was considered properly simulated.

RZWQM did not include routing capabilities for drainage flow. This limitation was not an important process for the soil column experiments, as the drain pipe length in the soil column was 28 cm and changes in the shape, peak, and time to peak of the subsurface hydrograph were considered insignificant. This may be an important limitation in the model as the contributing area being modeled increases.

SURFACE HYDROLOGICAL RESPONSE

For the plot experiments, the saturated hydraulic conductivity and soil structure macroporosity were the most important parameters for simulating the observed surface runoff (tables 2 and 3) during calibration. Note that biopores were not considered during the plot simulations. RZWQM could not properly represent the sub-event surface runoff (e.g., shape of the hydrograph, peak runoff, and runoff breakthrough time) as the model simulated surface runoff based on a one-dimensional process based exclusively on available excess water after infiltration (e.g., validated NSI of 0.01 and NOF of 0.55; table 5 and fig. 3). Routing, surface water storage, and exfiltration are not included. Deviations in the simulated runoff proportionally increased as a function of the

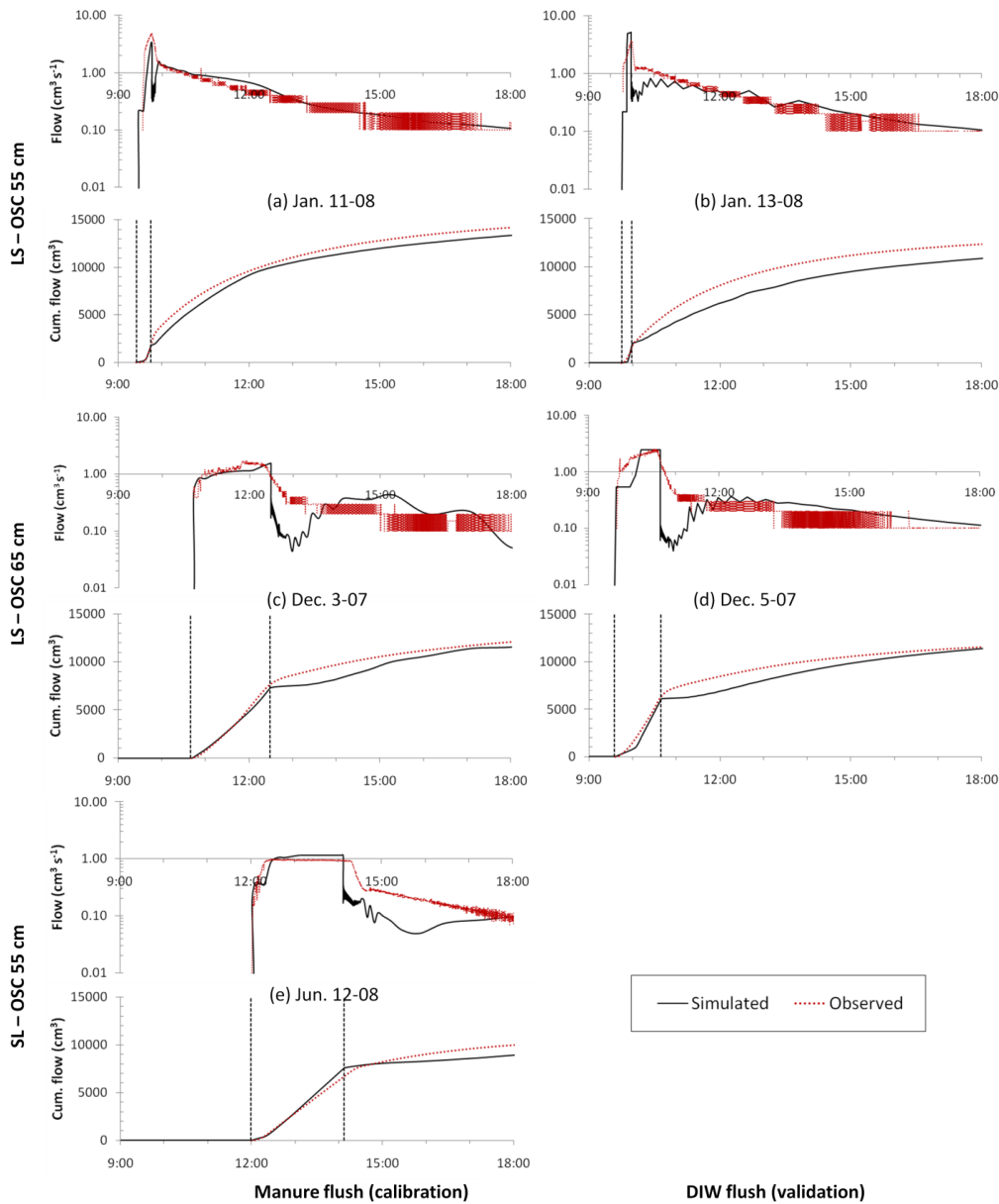


Figure 2. Subsurface drain flow and mass curve for the soil column experiments. Dashed vertical lines in the mass curve indicate the beginning and end of the inflow water event, and the horizontal axis is daily time.

rainfall intensity (NOF of 0.25 and 0.55 for the two-year and five-year storm events, respectively; table 5) due to an increase in the available surface runoff subjected to other surface processes. However, note that the total surface runoff volume and time to peak were properly estimated (mass curve errors below 2%; table 6 and fig. 3), providing reasonable estimations of surface runoff at the event scale. Moreover, the recession part of the hydrograph was absent in all simulations following the end of the rainfall event, as RZWQM considered surface runoff as lost from the system. This limitation may also play an important role in subsurface flow simulations with biopores, as available surface runoff cannot rapidly propagate through biopores during the recession part of the hydrograph.

SUBSURFACE FECAL BACTERIA TRANSPORT

The new transport routines and biopore concept allowed surface and subsurface transport simulation of fecal bacteria in surface runoff, subsurface drainage, soil matrix, macropores, and biopores with minimum calibration (table 6). Simulated EMCs in the soil column using the biopore concept were predicted with an absolute error in the range of 4% to 109%, NOF in the range of 0.36 to 3.34, and load errors in the range of 13% to 856% (fig. 4, table 7). In general, the predicted fecal bacteria concentrations and loads were on the same order of magnitude as the observed values. The large error of 856% in the estimated load for the loamy sand OSC-65 experiment (table 7) was explained by the lack of long-term measurements of fecal bacteria concentrations after the end of the distilled water flush. The NOF indicated high variability in simulated values with respect to observed measurements, as expected from the rapid transport typically

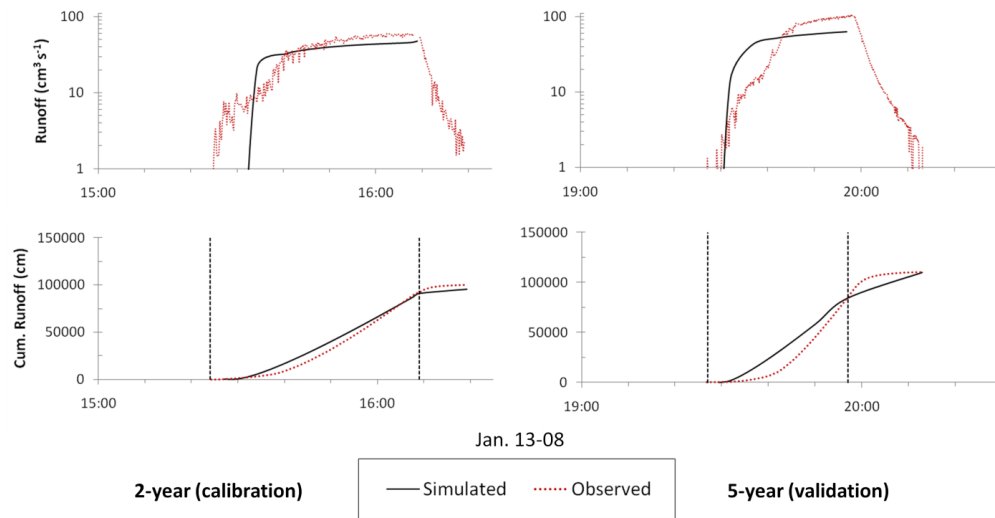


Figure 3. Surface runoff and mass curve for the plot experiments. Dashed vertical lines in the mass curve indicated the beginning and end of the artificial rainfall event, and the horizontal axis is daily time (27 May 2009).

Table 5. Surface runoff plot experiments calibration and validation results.

Experiment	Time ^[a]	Rainfall (cm)	Results ^[b]						
			MC _{obs} (cm ³)	MC _{sim} (cm ³)	NSI	NOF	MC _{err} (%)	Losses _{avg} (cm)	PIWEC (MPN mL ⁻¹)
Two-year storm	Start	15:24	4.38	0			--		
	Rainfall	16:09		90,949	90,955	0.81	0.25	0.0	1,660
	End	16:19		101,453	95,444			5.9	
Five-year storm	Start	19:27	3.81	0			--		
	Rainfall	19:57		85,460	84,471	0.05	0.55	1.2	1,660
	End	20:13		110,053	110,136			0.1	

^[a] “Rainfall” indicates the final time for the artificial rainfall event, and “End” indicates the final time for undetectable surface runoff.

^[b] MC_{obs} and MC_{sim} are the mass curve for observed and simulated surface runoff; NSI is the Nash-Sutcliffe efficiency index; NOF is the normalized objective function; MC_{err} is the mass curve error; Losses_{avg} is the average infiltration, interception, and evaporation volume; and PIWEC is the proportional inflow water *E. coli* concentration.

Table 6. Transport parameters for the soil column and plot experiment simulations.^[a]

Soil Type	Attachment			Die-off Rate (h ⁻¹)			
	A	b	K _{str}	D _{det}	LP	SP	Top
Loamy sand	0.38	1.010	0.01	0.24	0.05	0.03	0.1
Sandy loam	213.4	0.324	0.05 ^[b]	0.20	0.05	0.03	0.1
Silt loam	342.5	0.315	0.01 ^[b]	0.20 ^[b]	0.05	0.03	0.1

^[a] K_{str} is the straining coefficient, D_{det} is the detachment coefficient, LP is the liquid phase, SP is the solid phase, and Top represents the first 1 cm layer at the surface.

^[b] Calibrated parameter.

observed in the presence of macropores and biopores. This high variability resulted from the uncertainty associated with the bacteria quantification method, homogeneous (i.e., model) versus heterogeneous (i.e., soil column experiments) fecal bacteria input concentrations (note that experiments were conducted using diluted swine manure in which bacteria are mainly found attached to colloids rather than suspended free cells; Guzman et al., 2012), improper flow and fecal bacteria transport simulation from contributing soil layers, and rapid transport occurring due to biopore-macropore-drainage interconnectivity. During the inflow water event, the model was in good agreement with observations for calibration (note that in some cases model metric indexes were estimated based only a few measurements), but higher deviations were present during validation (fig. 4). On the other hand, for the

final depletion portion of the subsurface hydrograph, the model underestimated simulated fecal bacteria concentrations during calibration and overestimated concentrations during validation even though flow simulations were properly represented. These conditions were hypothesized to result from differences between the simulated and actual fecal bacteria mass at the above adjacent layers contributing most of the drainage flow in the model.

The soil column experiments resulted in different initial fecal bacteria conditions during calibration and validation. During calibration (e.g., manure flush), fecal bacteria concentrations were provided in the inflow water, and fecal bacteria pools in the soil profile (fig. 1) were initially zero. On the other hand, during validation (e.g., distilled water flush), the initial fecal bacteria along the soil profile pools (e.g., 1 cm soil layer thickness) were not zero. Therefore, the simulated fecal bacteria concentrations at the drain pipe were controlled by different transport processes during the manure and distilled water flush. During the manure flush, bacteria concentrations in the inflow water and downward transport to the end of the biopore mainly controlled simulated drain-age bacteria concentrations.

Note that transport through biopores occurred in RZWQM once surface runoff became available after infiltration through the soil and macropore domains. On the other hand, during the distilled water flush, available fecal bacteria at the lower soil layer pools above the drain pipe controlled drain-

Table 7. Fecal transport event mean concentrations (EMC), loads, and normalized objective function (NOF) for the soil column experiments; SNB refers to the simulations with no biopore routine included, and PIWEC is the proportional inflow water *E. coli* concentration.

Experiment	Flush	Time ^[a]	EMC (MPN mL ⁻¹)			EMC Error (%)	Loads (MPN)		Load Error (%)	NOF	PIWEC (MPN mL ⁻¹)
			Obs.	Sim.	SNB		Obs.	Pred.			
Loamy sand OSC-55	Manure	Inflow	6.5	13.6		109	4,820	13,300	176	3.34	153.6
		Total	22	6.9	0	68	354,000	76,900	78		
	Distilled water	Inflow	25	11.8		53	49,500	21,800	56	1.78	0
		Total	18	17.3	0	4	227,000	198,000	13		
Loamy sand OSC-65	Manure	Inflow	15	12		22	104,000	76,000	27	0.36	115.2
		Total	16	7.6	0	53	143,000	86,900	39		
	Distilled water	Inflow	14	26.5		88	89,900	157,000	75	1.38	0
		Total	14	17.6	0.3	28	95,100	909,000	856		
Silt loam OSC-55	Manure	Inflow	0.6	0.8		27	2,660	5,580	110	0.81	83.6
		Total	1.4	0.7	0	50	27,200	5,580	79		

[a] "Inflow" is the final time for the inflow event, and "Total" is the final time for the mass transport assessment.

age bacteria concentrations as a function of transport from the corresponding continuum volume (i.e., defined by the soil layer volume separating two drain pipes in RZWQM). The K_{HVF} parameter (i.e., effective horizontal layer volume contributing fecal bacteria in eq. 5) served to mimic physical non-equilibrium (e.g., transport occurring from a fraction of the continuum volume) at the horizontal layer (e.g., quasi two-dimensional flow and transport model using the integrated Hooghoudt's equation) occurring from fecal bacteria that propagated at each side of the biopore during the inflow event. For example, the K_{HVF} parameter ranged between 0.06 and 0.12 as a function of the layer depth, which represented 3.0 to 6.0 cm of the horizontal layer (note that the layer width was 50 cm) contributing fecal bacteria mass.

SURFACE RUNOFF FECAL BACTERIA TRANSPORT

For the plot experiments, concentrations as well as loads were underestimated in most cases (fig. 5, table 8). However, when using the upper poultry litter fecal bacteria concentrations reported by Guzman et al. (2010) (i.e., 2.6 times higher than the average *E. coli* concentration), the estimated errors in the simulated EMC and load were 38% and 46% for the two-year and five-year storm events, respectively. Note the significance of the fecal bacteria concentrations in the inflow

water event (rainfall or runoff) in the simulated runoff fecal bacteria concentrations (calibration and validation; fig. 5). This result was also recently reported by Fox et al. (2011) from a series of 2 m × 1 m laboratory-scale soil column experiments subjected to runoff events spiked with *E. coli*. They reported only small reductions in peak and event mean runoff concentrations, respectively. Infiltration also played an important role in transporting fecal bacteria loads through the soil profile and therefore reducing loads in surface runoff.

The NOF for the average simulated concentrations (table 8) indicated the presence of large deviations during calibration and validation; however, the simulated and observed values were on the same order of magnitude (fig. 5). The shape of the simulated fecal bacteria concentrations (i.e., breakthrough plateau in fig. 5) was properly represented for both calibration and validation. Aside from the variability due to an increase in available surface runoff, fecal bacteria concentrations are highly variable when litter is dispersed in water. Fecal bacteria in animal waste are mainly attached to colloids rather than free cells (Guzman et al., 2012), and transport of colloidal fecal bacteria may impact fecal concentrations in runoff. Future research is needed to incorporate routing processes in RZWQM and fecal bacteria release from manure application.

Table 8. Fecal transport event mean concentrations (EMC) and loads for the plot experiments, where PIWEC is the proportional inflow water *E. coli* concentration.

Experiment, Water Quality and the Propagation of Water in the Substation											
		Simulated			Error (%)			PIWEC (MPN mL ⁻¹)			
	Observed	Avg.	Min.	Max.	Avg.	Min.	Max.	NOF	Avg.	Min.	Max.
Two-year experiment											
EMC (MPN mL ⁻¹)	1,838	415	62	1,088	77.4	96.6	40.8	0.63	1,660	246	4,317
Load (MPN)	1.8 × 10 ⁸	3.8 × 10 ⁷	5.6 × 10 ⁶	9.9 × 10 ⁷	79.3	96.9	45.6				
Five-year experiment											
EMC (MPN mL ⁻¹)	1,784	446	67	1,169	75.0	96.2	34.5	0.71			
Load (MPN)	1.9 × 10 ⁸	3.8 × 10 ⁷	5.6 × 10 ⁶	9.9 × 10 ⁷	9.9	97.0	47.2				

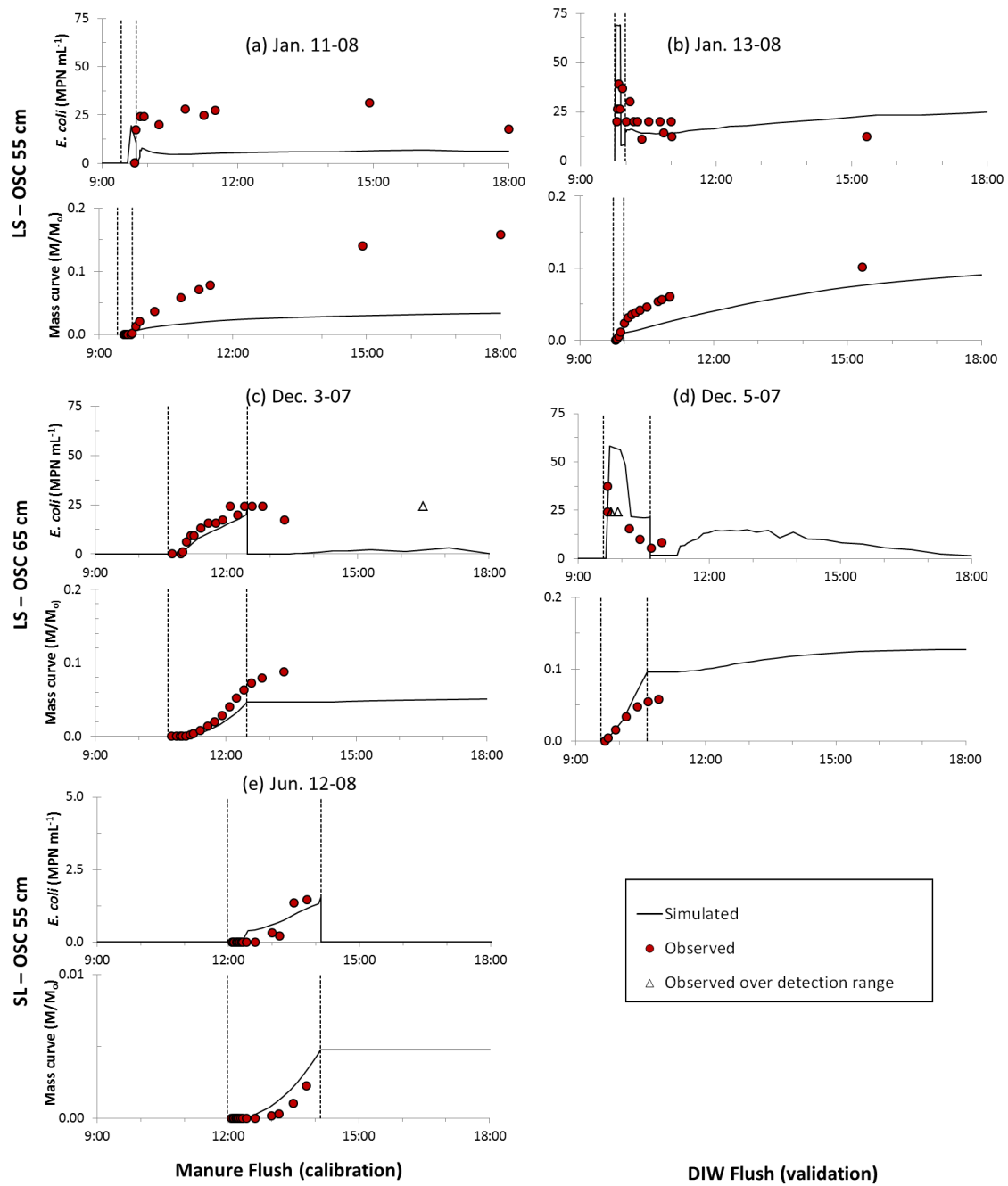


Figure 4. Observed and simulated fecal bacteria concentrations and mass curve for the soil column experiments. Dashed vertical lines indicated the beginning and end of the inflow event.

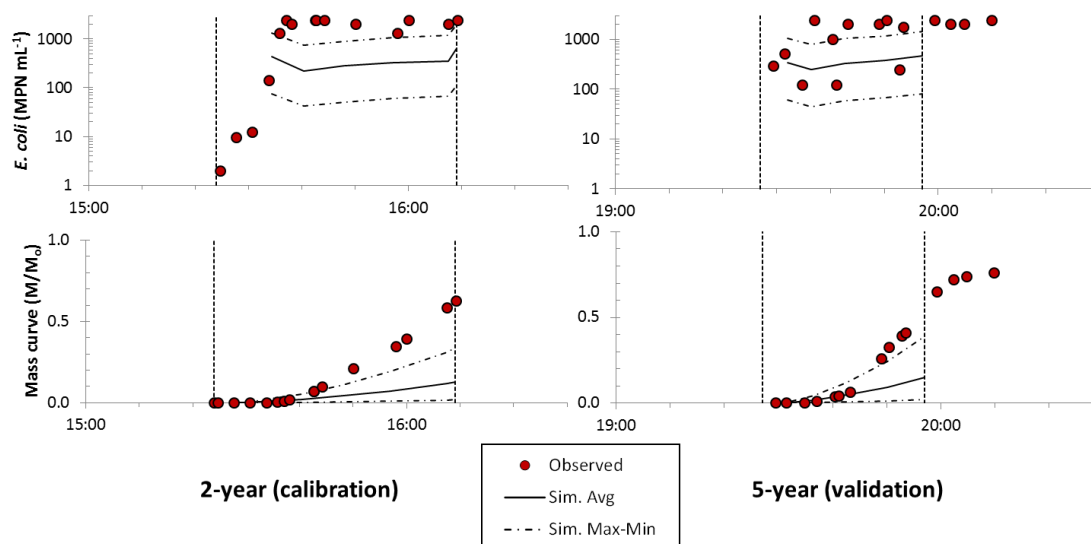


Figure 5. Observed and simulated fecal bacteria concentrations and mass curve for the plot experiments. Dashed vertical lines indicated the beginning and end of the artificial rainfall event.

SUMMARY AND CONCLUSIONS

The implementation of routines to simulate fecal bacteria transport in the vadose zone, as well as flow and transport through biopores, improved the capabilities of RZWQM to represent the rapid increase in flow and transport observed in subsurface drainage systems following rainfall/irrigation events. RZWQM was updated to simulate fecal bacteria transport in surface runoff. For model evaluations based on subsurface drainage experiments, the new routines improved the simulation of the subsurface hydrograph shape, time to peak, and flow breakthrough when biopores were present. In addition, fecal transport concentrations were properly represented in most cases. Evaluation of the model based on surface runoff experiments with artificial rainfall indicated that RZWQM can properly represent the total runoff volume but not the shape of the hydrograph, as RZWQM does not include surface runoff routing processes.

RZWQM is a lumped, physically based model that simulates a one-dimensional vertical soil profile with an integrated subsurface drainage component. RZWQM may not properly simulate cases with spatial variability in the input variables at any spatial scale. Therefore, the biopore concept can be successfully applied to those spatial scales in which the input variables can be considered homogenous. Implementation of routing processes for surface runoff and subsurface drain flow will improve RZWQM's capabilities for predicting flow and soil fecal bacteria transport at the field scale on subdaily time scales. Other facilitated transport processes, such as open surface disconnected biopores, should be incorporated into the model. Extensive research is needed to characterize biopores in field conditions and to help develop parameter inputs for fecal bacteria fate and transport immediately after manure applications and for simulating fecal bacteria population dynamics in the soil environment.

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